DAVPUBLICSCHOOLS,MPZONE-D

HALF YEARLY EXAMINATION

Session:2025-26

Class:X,Subject:AI

Marking Scheme

1. This Question Paper consists of **21 questions** in two sections: Section A & Section B.
2. Section A has Objective type questions whereas Section B contains Subjective type questions.
3. **Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours**.
4. All questions of a particular section must be attempted in the correct order.
5. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
   i. This section has 05 questions.
   ii. Marks allotted are mentioned against each question/part.
6. **SECTION B – SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
   i. This section has 16 questions.
   ii. A candidate has to do 10 questions .

(1x 20 = 20)

1.i(A) Ravish learns from the feedback and makes his project work better.
ii.Listener
iii. neat and clean appearance
iv.(c) Message through friend
v.d) All of the above mentioned
vi.(b) Microphone
2.i(d) All of these
ii.Artificial Intelligence (AI)
iii.(c) Who? What? Where? Why?
iv.(b)Statistical Data is one of the domains of AI model
v.(c) Data scraping
vifalse
3.i. (b) Self-Driving Car
ii.(a)Both A and R are correct and R is the correct explanation of A
iii(b) Alan Turing
iv)(c) Problem Scoping -> Data Acquisition -> Data Exploration -> Modelling -> Evaluation-> Deployment
v) (c) Graphical DI
vi) Numerical.

4.i.(d) All of these
ii.Computer vision
iii.(c) Both Statement 1 and Statement 2 are correct.
iv.(d) Textual, Tabular, Graphical
v.(c) Categories
vi.(c) Draw conclusions from data

5.i.(a) Find largest number
ii.(d) NLP
iii.(b) Emoji Scavenger
iv. (a) Discrete data
v.(b)Data
vi.(b)Python

6.The communication cycle is a fundamental concept that explains how information flows between individuals. It involves several key elements that ensure the message is sent, received, and understood effectively.

🔄 **Elements of the Communication Cycle**

1. **Sender**
   - The person who initiates the message.
2. **Message**
   - The actual information, idea, or feeling being communicated.
3. **Encoding**
   - The process of converting the message into symbols, language, or gestures.
4. **Medium/Channel**
   - The method used to transmit the message (e.g., speech, email, text, video).
5. **Receiver**
   - The person who gets the message and decodes it.
6. **Decoding**
   - The receiver interprets and understands the message.
7. **Feedback** 🔁
   - The receiver responds, confirming whether the message was understood correctly.

7. Elements of a Communication Process

i. Sender (Communicator)

- The originator of the message.

ii. Message

- The content of the communication — what the sender wants to convey.

iii. Encoding

- The process of converting thoughts into a communicable format**.**

iv. Medium (Channel)

- The method or pathway used to transmit the message.

v. Receiver

- The person or group who receives and decodes the message.

vi. Decoding

- The process by which the receiver interprets or makes sense of the message.

vii. Feedback

- The receiver's response to the message.

8. A person is called an entrepreneur not because they own a big business, but because they start something new, take risks, and innovate. Therefore, the statement is not always true

9.1. Improved Communication

ICT tools (like mobile phones, email, social media) allow instant and cost-effective communication with anyone around the world.

2. Better Access to Information

The internet helps individuals access real-time information on news, health, education, government schemes, weather, etc.

3. Convenience in Daily Tasks

ICT enables online shopping, e-banking, e-ticketing, and utility payments, saving time and effort.

10. Self-confidence is the belief in one's own abilities, qualities, and judgment . It reflects how much trust a person has in themselves to handle challenges, make decisions, and pursue goals.

11.. Voice Assistants

- Examples: Alexa, Siri, Google Assistan

Recommendation Systems

- Examples: YouTube, Netflix, Amazon

Face Recognition

- Used in smartphones for unlocking and in security systems

12Key Features of Python:

1. Simple and Easy to Learn
   - Python has easy-to-read syntax, making it ideal for beginners.
2. Interpreted Language
   - Code is executed line by line, which makes debugging easy.
3. High-Level Language
   - Python is close to human language, hiding complex low-level details.
4. Open Source and Free

- Python is freely available and supported by a large community.

5. Portable
   - Python programs can run on different platforms (Windows, Mac, Linux) without changes.
6. Extensive Libraries
   - Python offers rich libraries like NumPy, Pandas, Matplotlib, TensorFlow, etc., for various tasks.
7. Object-Oriented :
   - Supports both object-oriented and procedural programming styles.

13. The principles of AI ethics are guidelines that ensure Artificial Intelligence is developed and used responsibly, fairly, and safely. These include:

1. Fairness
AI systems should treat everyone equally and without bias, regardless of gender, race, age, etc.
2. Transparency
How AI works and makes decisions should be clear and understandable to users.
3. Accountability
Humans must be responsible for the actions and outcomes of AI systems.
4. Privacy
AI should protect users' personal data and use it responsibly with consent.
5. Safety and Security
AI systems must be safe to use and protected from misuse or cyberattacks.
14. Understand basic concept of Data:
– Learn about numbers, text, dates, and categories.
Learn to read graphs and charts:
– Practice reading bar graphs, pie charts, line graphs, etc.
Practice Data Collection and organization
– Use surveys or spreadsheets to collect and organize data.
Use Tools Like Excel or Google Sheets
– Learn basic data functions, sorting, filtering, and formulas.
Interpret and analyse data:
– Try to find patterns or trends in the data.
15.Qualitative

- Reason: Customer feedback consists of descriptive comments like "Great quality!" or "Could be better." These are non-numeric and express opinions, experiences, or sentiments—hallmarks of qualitative data.

2. Continuous
   - Reason: Ratings like 4.5, 4.0, 3.8 are decimal values, which means they can take on any value within a range (e.g., 1.0 to 5.0). This makes them continuous data, as opposed to discrete data which only includes whole numbers.

16. Best Practices of Cyber Security
Do not share personal information
Install antivirus and keep software updated.
17.1. Quantitative Data vs Qualitative Data

Quantitative Data	Qualitative Data
Refers to numerical data	Refers to descriptive or categorical data
Example: Height, age, marks, income	Example: Color, gender, feedback, emotions
Can be measured and analyzed statistically	Can be observed but not easily measured

Primary Data Secondary Data

Collected **first-hand** by the researcher Already collected and available from other sources

Original and specific to the purpose May not be specific to current needs

Example: Surveys, interviews, experiments Example: Websites, reports, books, databases

Data Privacy **Data Security**

Focuses on **who is allowed to access the data** Focuses on **protecting data from unauthorized access**

Ensures personal data is used lawfully Ensures data is safe from hackers and loss

Example: Consent before data collection Example: Use of passwords, encryption, firewalls

18.i. The Data Pyramid is a conceptual model that illustrates how raw data is transformed into meaningful insights and ultimately wise decisions. It's widely used in fields like data science, education, and management to show the progression from basic facts to deep understanding.

- It has four levels: Data → Information → Knowledge → Wisdom
- Each level adds context, meaning, and value to the one below it

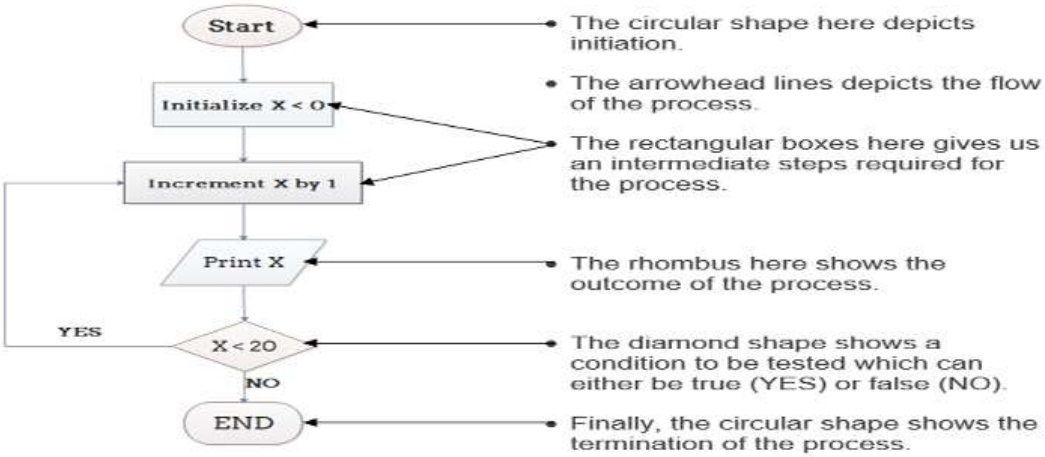
ii.

Label	Level
D	**Data**
C	**Information**
B	**Knowledge**
A	**Wisdom**

iii.

Level	Feature
Data (D)	Raw facts and figures without context. Examples: numbers, dates, symbols.
Information (C)	Processed data that answers basic questions like "who," "what," "when," "where."
Knowledge (B)	Insights gained by analyzing and connecting information. Explains "how."
Wisdom (A)	Deep understanding that guides decision-making. Answers "why" and "what should be done."

20.Algorithm: Step 1: Initialize X as 0,
Step 2: Increment X by 1,
Step 3: Print X,
Step 4: If X is less than 20 then go back to step 2.
Flowchart:



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